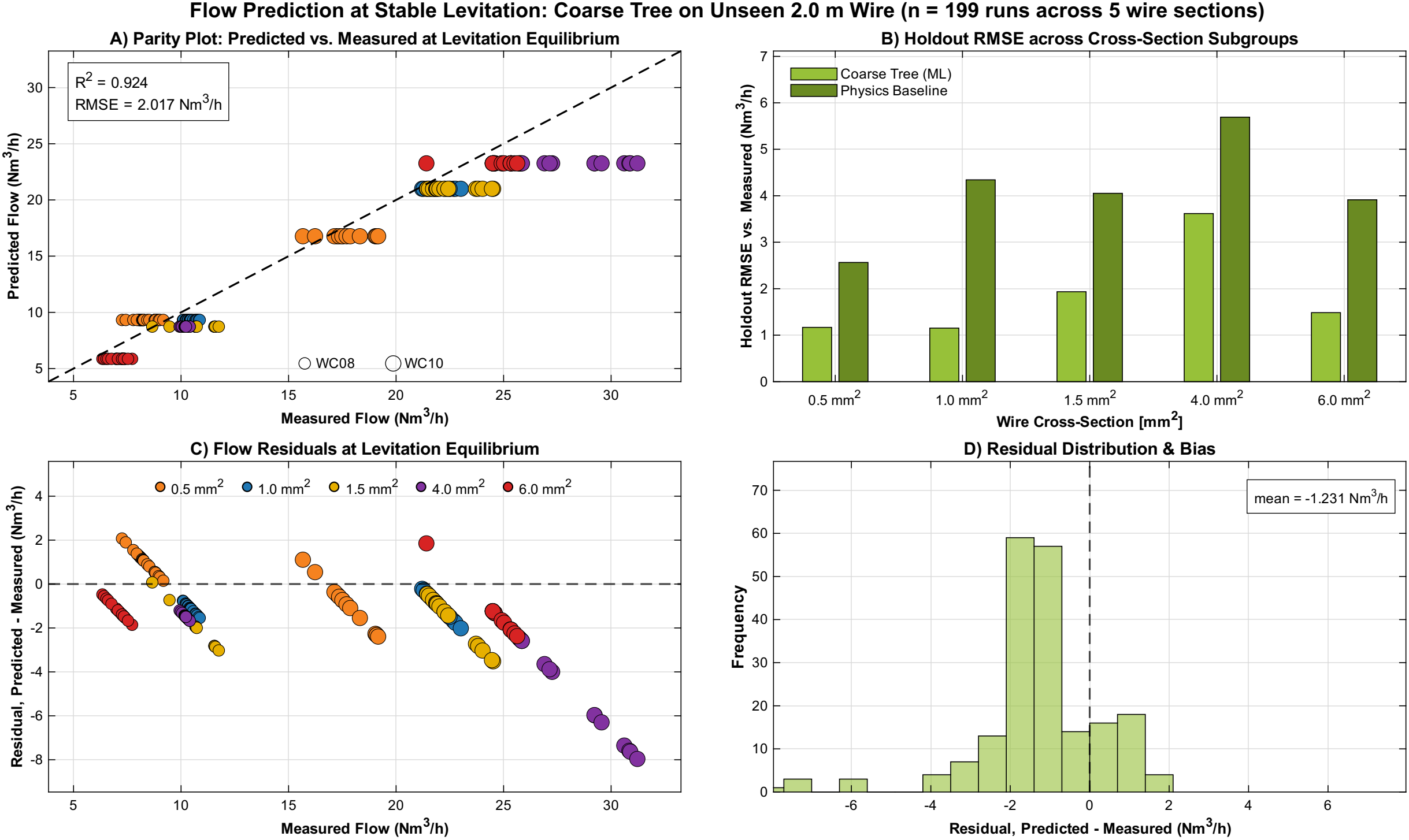

Flow Prediction at Stable Levitation: Coarse Tree on Unseen 2.0 m Wire (n = 199 runs across 5 wire sections)
A) Parity Plot: Predicted vs. Measured at Levitation Equilibrium
R2 = 0.924
RMSE = 2.017 Nm3/h
WC08
WC10
Predicted Flow (Nm3/h)
Measured Flow (Nm3/h)
B) Holdout RMSE across Cross-Section Subgroups
Coarse Tree (ML)
Physics Baseline
Holdout RMSE vs. Measured (Nm3/h)
0.5 mm2
1.0 mm2
1.5 mm2
4.0 mm2
6.0 mm2
Wire Cross-Section [mm2]
C) Flow Residuals at Levitation Equilibrium
0.5 mm2
1.0 mm2
1.5 mm2
4.0 mm2
6.0 mm2
Residual, Predicted - Measured (Nm3/h)
Measured Flow (Nm3/h)
D) Residual Distribution & Bias
mean = -1.231 Nm3/h
Frequency
Residual, Predicted - Measured (Nm3/h)